FOOD DONATION TRACKER

A Python and MySQL project connecting surplus food with NGOs

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1. Introduction

The Food Donation Tracker is a Python and MySQL program that lets hotels, caterers and households post surplus food, and lets NGOs claim it before it is wasted. It replaces scattered phone calls and messages with one shared record of who donated what, who collected it, and how many meals were saved.

The project has two layers. MySQL is the **back end**: it stores the data in three linked tables called Donors, NGOs and Donations. Python is the **front end**: it shows a menu, takes the user's input, and sends SQL queries to the database through the mysql.connector module. It uses topics from the Class XII Computer Science course: Python functions and loops, SQL queries with joins and aggregate functions, and Python–SQL connectivity.

2. Need for the Project

2.1 Food is wasted while people go hungry

The United Nations Environment Programme (UNEP) estimated that the world wasted 1.05 billion tonnes of food in 2022. That is about one fifth (19 per cent) of all the food available to consumers, and it happened in a year when 783 million people were hungry (UNEP, 2024a).

Most of this waste does not happen on farms. Households were responsible for 60 per cent of it, food services such as hotels and restaurants for 28 per cent, and retail shops for 12 per cent (UNEP, 2024b). The same report calculated that households alone waste at least one billion meals every day (UNEP, 2024a).

India is a large part of this picture. Its food waste was about 78 million tonnes in 2022, the second highest in the world after China, and the average person discarded 55 kg of food at home that year, compared with 73 kg in the United States (Statista, n.d.). The per-person figure looks modest, but India's large population makes the total very high.

Hunger remains a concern at the same time. India ranked 100th among 122 countries in the Global Hunger Index 2026, with a score of 24.1, which falls in the “serious” category (Careers360, 2026). The index combines four measures (undernourishment,

child stunting, child wasting and child mortality), so it is quoted here only to show that hunger has not disappeared, not as a count of people without food.

Wasted food also harms the environment. Food loss and waste produce 8 to 10 per cent of global greenhouse gas emissions (UNEP, 2024a).

2.2 The missing link between donors and NGOs

A wedding hall, a hotel buffet or a hostel mess often ends the day with cooked food left over. Cooked food cannot be kept for long, so the donor has to find someone to collect it quickly. NGOs face the opposite problem. They do not know where surplus food is available on a given day, and a single phone call may reach an NGO that has no vehicle or volunteers free. The food is wasted not because nobody wants it, but because the two sides cannot find each other in time.

India's food regulator has already recognised this gap. In 2019 the Food Safety and Standards Authority of India (FSSAI) notified the Food Safety and Standards (Recovery and Distribution of Surplus Food) Regulations, 2019, to protect organisations and individuals who donate food in good faith and to encourage donation to non-profit organisations (FSSAI, 2019). The regulations came into force when they were published in the Gazette on 26 July 2019, and food businesses had to comply by 1 July 2020 (Packaging South Asia, 2019). They also list records that must be kept, such as the donor and distributor names, the donation date, the product name and the quantity donated (Packaging South Asia, 2019). FSSAI also formed the Indian Food Sharing Alliance to bring together food recovery agencies and NGOs (FSSAI, n.d.).

What is still needed at the local level is a simple tool that keeps this record and puts donors and NGOs in touch. The Food Donation Tracker is a small prototype of such a tool.

2.3 Why a database program

The information in this problem has to be stored, searched, updated and summarised: which donors exist, which food is still available, which NGO took it, and how many meals were saved each month. A database does all four reliably, and several people can use the same data. Python then gives the database a friendly front end, so that a person who knows no SQL can still use it.

3. Objectives

- To let donors register and post surplus food with the item name and the number of meals.
- To let NGOs register, see all available food in one list, and accept a pickup.
- To make sure one donation can be claimed by only one NGO.
- To keep a permanent record of every donation with its date, donor and collecting NGO.

- To report the total meals saved in each month.
- To apply Python–MySQL connectivity to a real-world problem.

4. What It Can Be Used For

The program is a prototype, but the same design suits many places where cooked or packaged food is left over regularly.

Where	How it would be used
Hotels, restaurants and caterers	Post the unsold food at closing time. A nearby NGO accepts it, and the hotel gets a dated record of the donation.
Marriage halls and event venues	Post the extra food right after a function, when it is still fresh and the donor wants it collected quickly.
Canteens and hostel messes	School, college or hostel kitchens record surplus meals and hand them to a registered NGO.
Housing societies and offices	A community or office cafeteria posts leftovers from festivals and large events.
NGOs and food banks	See every available donation in one place, claim it, and use the monthly report to show their work to supporters.
Volunteer groups and local bodies	Track how many meals are rescued in an area each month.
Classrooms	A worked example of a front end and a back end for teaching Python–SQL connectivity.

Note: these are possible uses. Before real use, the program would need the safety and security improvements listed in Section 8, especially a record of when the food was cooked and when it expires.

5. Benefits

Who	Benefit
Donors	Posting takes a minute. The donor gets a written record of the date, the item and the number of meals, which fits the record-keeping idea in the FSSAI regulations. Food that would have been thrown away reaches people.
NGOs	One list of all available food, with the donor's name, phone number and city. The first NGO to accept gets the food, and the program refuses a second claim, so two NGOs do not travel to collect the same donation.
Society	More meals reach people in need. Less food goes to waste, and less waste means fewer greenhouse gases, since food loss and waste cause 8 to 10 per cent of the world's emissions (UNEP, 2024a).

Who	Benefit
Organisers and supporters	The monthly report gives a clear number: meals accepted by NGOs in each month. It measures meals rescued for distribution, not meals eaten.
The student	A complete working project that joins the Python, SQL and connectivity chapters, solves a real problem, and can be extended with new features.

6. How the System Works

6.1 Design

The user never touches the database directly. Each menu choice is handled by a Python function, which builds an SQL query, sends it to MySQL with the cursor, and prints the result in plain words (Figure 1).

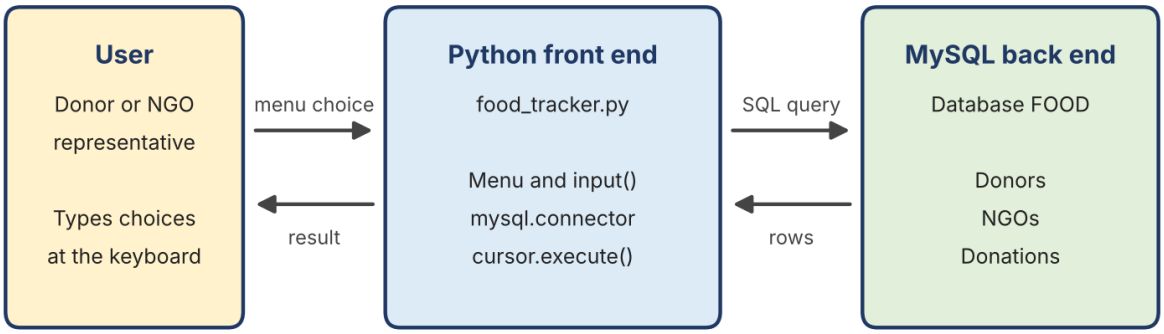


Figure 1: Python front end and MySQL back end

A donation passes through a short life cycle. It starts as Available when the donor posts it, and changes to Accepted when an NGO claims it. The reports use the Accepted records (Figure 2).



Figure 2: Life of a donation

6.2 Database tables

The database is named FOOD. Donors and NGOs each have one row per organisation, and every row in Donations points to one donor and, once accepted, to one NGO through the ID columns.

Table	Column	Type	Purpose
Donors	donor_id	INT, primary key	Automatically numbered ID of the donor
	donor_name	VARCHAR(50)	Name of the hotel, caterer or person
	donor_phone	VARCHAR(10)	Contact number
	donor_city	VARCHAR(30)	City of the donor
NGOs	ngo_id	INT, primary key	Automatically numbered ID of the NGO
	ngo_name	VARCHAR(50)	Name of the NGO
	ngo_phone	VARCHAR(10)	Contact number
Donations	donation_id	INT, primary key	Automatically numbered ID of the donation
	donor_id	INT	Which donor posted it (matches Donors)
	ngo_id	INT	Which NGO accepted it (matches NGOs); empty until accepted
	food_item	VARCHAR(50)	What the food is
	meals	INT	Number of meals
	donate_date	DATE	Date of posting, filled in by CURDATE()
	status	VARCHAR(10)	Available (default) or Accepted

6.3 Menu options

Menu option	What it does	SQL idea used
1. Add Donor	Registers a donor and shows the new Donor ID	INSERT, AUTO_INCREMENT
2. Add NGO	Registers an NGO and shows the new NGO ID	INSERT
3. Post Food	Adds a donation dated today	INSERT, CURDATE()
4. View Available Food	Lists food not yet accepted, with donor contact details	SELECT, NATURAL JOIN, WHERE
5. NGO Accepts Pickup	Marks a donation as accepted by the NGO	UPDATE, WHERE, rowcount

Menu option	What it does	SQL idea used
6. Monthly Meals Report	Total meals accepted in each month	SUM, GROUP BY, ORDER BY
7. Pickup Details	Shows which NGO accepted which donation	NATURAL JOIN of three tables

Preventing double claims. The acceptance query is: UPDATE Donations SET ngo_id = …, status = 'Accepted' WHERE donation_id = … AND status = 'Available'. The condition status = 'Available' means that once one NGO has accepted a donation, the same query changes nothing for the next NGO. The program checks the number of rows changed (cur.rowcount) and tells the second NGO that the food is already taken.

6.4 A typical session

1. A hotel registers as a donor (option 1) and notes its Donor ID.
2. An NGO registers (option 2) and notes its NGO ID.
3. At closing time the hotel posts 40 meals of veg biryani (option 3).
4. The NGO opens the list of available food (option 4), sees the hotel's name, phone number and city, and accepts the donation (option 5).
5. At the end of the month, option 6 shows the meals saved and option 7 shows who collected what.

The complete working of one such session, as printed by the program, is given in Appendix B.

7. Class XII Concepts Used

The syllabus for Class XII Computer Science (2026–27) lists SQL with DDL and DML, ORDER BY, aggregate functions, GROUP BY, joins, and the Python interface functions connect(), cursor(), execute(), commit(), fetchone(), fetchall() and rowcount (Directorate of Education, GNCT of Delhi, 2026). Its project guidelines ask for something tangible and useful that solves a real-world problem using Python–SQL connectivity (Directorate of Education, GNCT of Delhi, 2026). The table shows where each idea appears in this project.

Concept	Where it is used in the project
DDL: CREATE DATABASE, CREATE TABLE	The program creates the FOOD database and the three tables on its first run.
DML: INSERT, UPDATE, SELECT	INSERT adds donors, NGOs and donations; UPDATE records an acceptance; SELECT feeds the lists and reports.

Concept	Where it is used in the project
Primary key, AUTO_INCREMENT, DEFAULT	Every table has a numbered ID; a new donation gets the status Available by default.
Natural join	Options 4 and 7 join Donations with Donors, and with NGOs, on the common ID column.
Aggregate function, GROUP BY, ORDER BY	The monthly report uses SUM(meals), grouped by year and month and sorted in date order.
Date functions	CURDATE(), YEAR() and MONTHNAME() supply and format the dates. These are a small extra beyond the syllabus list.
connect(), cursor(), execute(), commit()	Open the connection, send each query, and save the changes.
fetchall() and rowcount	fetchall() brings back the rows to print; rowcount tells whether an acceptance succeeded. The cursor's lastrowid, which gives the new ID after an INSERT, is another small extra.
Python basics	Functions for each menu option, a while loop for the menu, if–elif for the choices, input() and int(), and string formatting with format().

8. Limitations and Future Scope

8.1 Limitations

The program was tested with sample data, and the tests showed the following limits, which are normal for a first prototype.

- It runs in a text window on one computer, so donors and NGOs cannot use it from their own phones.
- The program trusts the IDs typed in. An ID that does not exist is not rejected, so an unknown NGO ID can be recorded as accepting food.
- Queries are built by inserting the typed text into the SQL string with format(). A name containing an apostrophe, such as O'Brien, makes the query fail, and on a public system this style is unsafe.
- A phone number longer than ten characters, or text typed where a number is expected, stops the program with an error.
- It does not record when the food was cooked or when it expires, so it cannot judge whether the food is still safe.
- There is no login, no alert to NGOs when food is posted, and no matching by location.

8.2 Future scope

- Pass values to MySQL as parameters (%s placeholders), add foreign keys, and check IDs and phone numbers before saving.

- Add cooked time, expiry time, batch and storage temperature, which also correspond to the record fields listed in the FSSAI regulations (Packaging South Asia, 2019).
- Add a Collected status, so that a pickup can be marked complete after the NGO reaches the donor.
- Build a window-based version with Tkinter, or a web or mobile version, so that real users can reach it.
- Send an SMS or WhatsApp message to NGOs in the donor's city when new food is posted.
- Draw monthly charts of meals saved with the matplotlib library.

9. Conclusion

Food is wasted every day in large quantities while hunger continues, and part of the reason is that surplus food and the organisations that can collect it do not meet in time. The Food Donation Tracker shows how a small Python front end and a MySQL database can close that gap: donors post food, NGOs claim it once, and the system keeps a clear record and a monthly count of meals saved.

The project also applied the main Class XII topics, from table design and joins to cursors and commits, to a problem that matters outside the classroom. With the improvements in Section 8, the same design could grow from a classroom prototype into a tool that a real neighbourhood could use.

10. References

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Appendix A: Source Code

File: food_tracker.py. Replace your_password_here with the MySQL root password before running. The database and tables are created automatically on the first run. Long lines wrap in this printout; in the program file each is a single line.

```
import mysql.connector
con = mysql.connector.connect(host="localhost", user="root", passwd="your_password_here")
cur = con.cursor()
cur.execute("CREATE DATABASE IF NOT EXISTS FOOD")
cur.execute("USE FOOD")
cur.execute("""CREATE TABLE IF NOT EXISTS Donors (
  donor_id   INT AUTO_INCREMENT PRIMARY KEY,
  donor_name  VARCHAR(50),
  donor_phone VARCHAR(10),
  donor_city  VARCHAR(30))""")

cur.execute("""CREATE TABLE IF NOT EXISTS NGOs (
  ngo_id   INT AUTO_INCREMENT PRIMARY KEY,
  ngo_name  VARCHAR(50),
  ngo_phone VARCHAR(10))""")

cur.execute("""CREATE TABLE IF NOT EXISTS Donations (
  donation_id INT AUTO_INCREMENT PRIMARY KEY,
  donor_id   INT,
  ngo_id     INT,
  food_item  VARCHAR(50),
  meals      INT,
  donate_date DATE,
  status     VARCHAR(10) DEFAULT 'Available')""")
con.commit()
def add_donor():
  name = input("Donor name: ")
  phone = input("Phone: ")
  city = input("City: ")
  q = "INSERT INTO Donors (donor_name, donor_phone, donor_city) VALUES ('{}', '{}', '{}')".format(name, phone,
city)
  cur.execute(q)
  con.commit()
  print("Donor added. Your Donor ID is", cur.lastrowid)
def add_ngo():
  name = input("NGO name: ")
  phone = input("Phone: ")
  q = "INSERT INTO NGOs (ngo_name, ngo_phone) VALUES ('{}', '{}')".format(name, phone)
  cur.execute(q)
  con.commit()
  print("NGO added. Your NGO ID is", cur.lastrowid)
def post_food():
  donor_id = int(input("Your Donor ID: "))
  item = input("Food item: ")
  meals = int(input("Number of meals: "))
  q = "INSERT INTO Donations (donor_id, food_item, meals, donate_date) VALUES ({}, '{}', {},
CURDATE())".format(donor_id, item, meals)
  cur.execute(q)
  con.commit()
  print("Food posted. Donation ID is", cur.lastrowid)
def view_food():
  cur.execute("""SELECT donation_id, donor_name, donor_phone, donor_city, food_item, meals
           FROM Donations NATURAL JOIN Donors
           WHERE status = 'Available'""")
  rows = cur.fetchall()
  if len(rows) == 0:
    print("No food available now.")
  for r in rows:
    print("ID:", r[0], "| Donor:", r[1], "| Phone:", r[2], "| City:", r[3],
      "| Food:", r[4], "| Meals:", r[5])

def accept_food():
  ngo_id = int(input("Your NGO ID: "))
  donation_id = int(input("Donation ID to accept: "))
```

```
  q = "UPDATE Donations SET ngo_id = {}, status = 'Accepted' WHERE donation_id = {} AND status =
'Available'".format(ngo_id, donation_id)
  cur.execute(q)
  con.commit()
  if cur.rowcount == 1:
    print("Pickup accepted. Please collect the food.")
  else:
    print("Wrong ID or food already taken.")
def monthly_report():
  cur.execute("""SELECT MONTHNAME(donate_date), YEAR(donate_date), SUM(meals)
          FROM Donations
          WHERE status = 'Accepted'
          GROUP BY YEAR(donate_date), MONTH(donate_date), MONTHNAME(donate_date)
          ORDER BY YEAR(donate_date), MONTH(donate_date)""")
  rows = cur.fetchall()
  if len(rows) == 0:
    print("No accepted donations yet.")
  for r in rows:
    print(r[0], r[1], ": total meals =", r[2])
def pickup_details():
  cur.execute("""SELECT donation_id, donor_name, ngo_name, food_item, meals
          FROM Donations NATURAL JOIN Donors NATURAL JOIN NGOs
          WHERE status = 'Accepted'""")
  rows = cur.fetchall()
  if len(rows) == 0:
    print("No completed pickups yet.")
  for r in rows:
    print("ID:", r[0], "| Donor:", r[1], "| NGO:", r[2], "| Food:", r[3], "| Meals:", r[4])
while True:
  print("\n===== FOOD DONATION TRACKER =====")
  print("1. Add Donor")
  print("2. Add NGO")
  print("3. Post Food")
  print("4. View Available Food")
  print("5. NGO Accepts Pickup")
  print("6. Monthly Meals Report")
  print("7. Pickup Details")
  print("0. Exit")
  ch = input("Enter choice: ")

  if ch == "1":
    add_donor()
  elif ch == "2":
    add_ngo()
  elif ch == "3":
    post_food()
  elif ch == "4":
    view_food()
  elif ch == "5":
    accept_food()
  elif ch == "6":
    monthly_report()
  elif ch == "7":
    pickup_details()
  elif ch == "0":
    print("Thank you!")
    break
  else:
    print("Invalid choice")

con.close()
```

Appendix B: Sample Run

Output from a test run with sample data (the names are invented). The month shown in the report is the month in which the test was run.

```
Enter choice: 1
Donor name: Asha Hotel
Phone: 9876543210
City: Coimbatore
Donor added. Your Donor ID is 1

Enter choice: 2
NGO name: Hope NGO
Phone: 9123456780
NGO added. Your NGO ID is 1

Enter choice: 3
Your Donor ID: 1
Food item: Veg biryani
Number of meals: 40
Food posted. Donation ID is 1

Enter choice: 3
Your Donor ID: 1
Food item: Chapati
Number of meals: 25
Food posted. Donation ID is 2

Enter choice: 4
ID: 1 | Donor: Asha Hotel | Phone: 9876543210 | City: Coimbatore | Food: Veg biryani | Meals: 40
ID: 2 | Donor: Asha Hotel | Phone: 9876543210 | City: Coimbatore | Food: Chapati | Meals: 25

Enter choice: 5
Your NGO ID: 1
Donation ID to accept: 1
Pickup accepted. Please collect the food.

Enter choice: 5
Your NGO ID: 1
Donation ID to accept: 1
Wrong ID or food already taken.

Enter choice: 4
ID: 2 | Donor: Asha Hotel | Phone: 9876543210 | City: Coimbatore | Food: Chapati | Meals: 25

Enter choice: 6
November 2026 : total meals = 40

Enter choice: 7
ID: 1 | Donor: Asha Hotel | NGO: Hope NGO | Food: Veg biryani | Meals: 40
```